ALeRCE
Automatic Learning for the Rapid Classification of Events

The ALeRCE Astronomical Broker: Services

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ALeRCE HackFest
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https://science.alerce.online

CMM Center for Mathematical Modeling
INSTITUTO MILENIO DE ASTROFÍSICA
DATA OBSERVATORY
UNIVERSIDAD ADOLFO IBÁÑEZ
UTEM
REUNA Ciencia y Educación en Red
aws
NLHPC National Laboratory for High Performance Computing Chile
IACS
DiRAC

ALeRCE public real-time products from stream data (as of today)

Service	ZTF	LSST
Explorer	https://alerce.online/	https://lsst.alerce.online/
SN Hunter	https://snhunter.alerce.online/	Not yet
Watchlist	https://watchlist.alerce.online/	Not yet
Python client	https://alerce.readthedocs.io/en/latest/	https://alerce.readthedocs.io/en/latest/
API	https://api.alerce.online/ztf/v1	See info here
Direct DB connection (SQL queries)	Yes	No
TAP access (ADQL queries)	Not yet	https://tap.alerce.online/
Stream	Upon request	Upon request
Use case jupyter notebooks	https://github.com/alercebroker/usecases/tree/master/notebooks/ZTF	https://github.com/alercebroker/usecases/tree/master/notebooks/LSST

ALeRCE public real-time products from ZTF data

Presentation paper (Förster+2021)

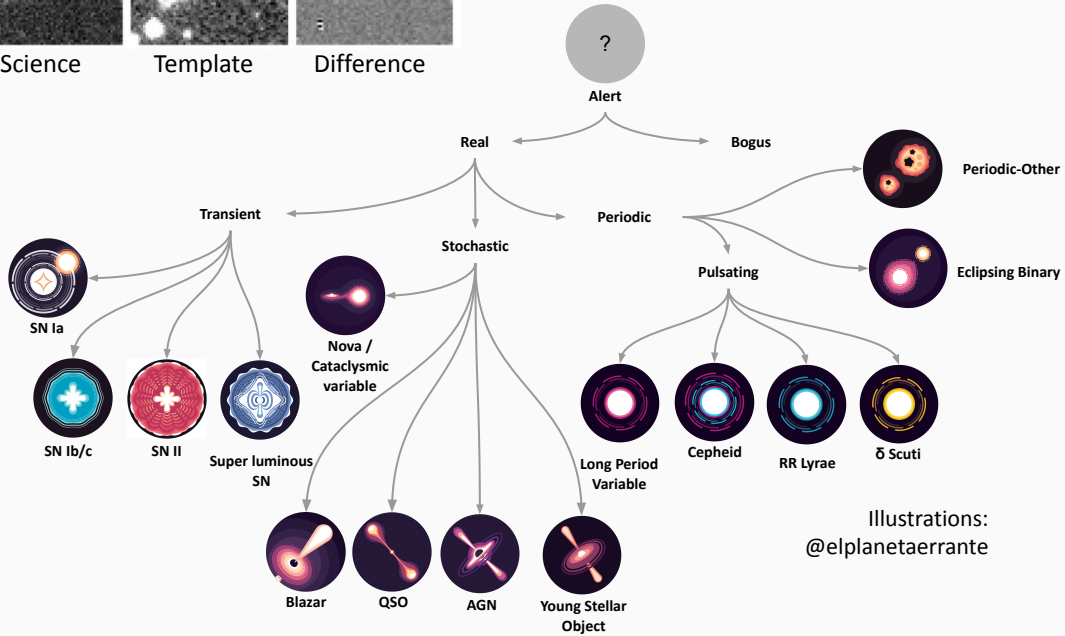
Stamp classifier (Carrasco-Davis+2021)

- Convolutional neural network
- 5 classes: SN, AGN, VS, Asteroid, Bog
- Uses first triplet of stamps per object
- Enables fast discovery of transient candidates

Light curve classifier (Sánchez-Sáez+2021)

- Balanced hierarchical random forest
- 15 classes
- Starts from 6 detections in any band per object
- Updates after every new detection
- Involves computing up to 174 features per object

Stream start: June 2018
>580 M detections processed
>5.1 B non-detections
>987 M forced photometry epochs
>129 M unique objects



Illustrations: @elplanetaerrante

ALeRCE classifiers for ZTF data

classifier_name (in 'probability' table)	classifier_version (latest)	Reference	Starting date	Latest feature version used
lc_classifier	23.12.26a85	Sánchez-Sáez+2021	~2019	23.12.26a85
lc_classifier_BHRF_forced_phot	2.1.0	Cabrera-Vives+ in prep.	May 29 2025	27.5.6
LC_classifier_ATAT_forced_phot(beta)	1.1.0_beta	Cabrera-Vives+ in prep. (but see also Cabrera-Vives+2024)	June 3 2025	27.5.6
stamp_classifier	stamp_classifier_1.0.4	Carrasco-Davis+2021	~2019	-
stamp_classifier_2025_beta	2.1.1_beta	In prep. (but see also Reyes-Jainaga+2023)	July 4 2025	-

ALeRCE public real-time products from LSST data

Science Template Difference

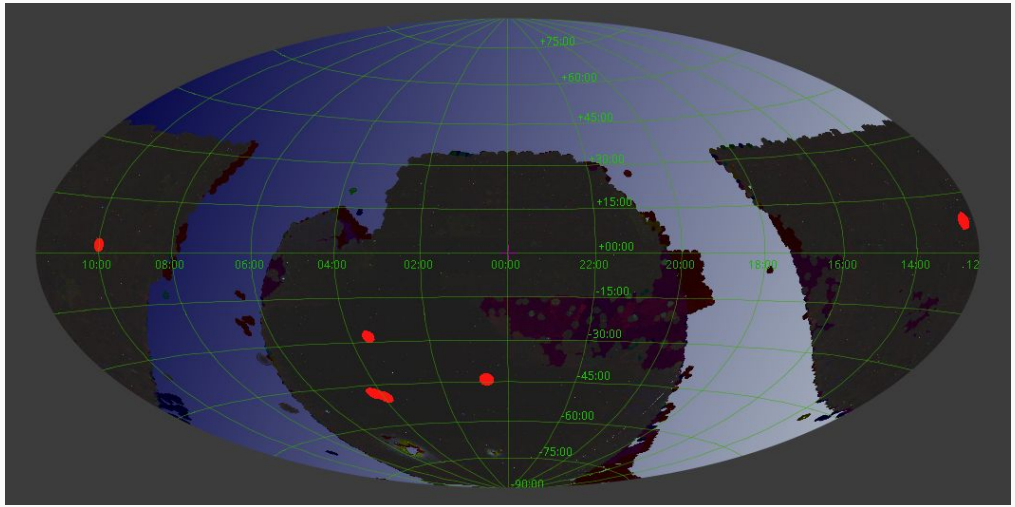
Stream start: Sept 2025
>11.3 M detections processed
>46 M forced photometry epochs
>1.6 M unique diaObjects
>1.3 k unique ssObjects

Stamp classifier (in prep.)

- Convolutional neural network
- 5 classes: SN, AGN, VS, Asteroid, Bogus
- Uses first triplet of stamps per object
- Enables fast discovery of transient candidates

Features

- Starts from 2 detections per object
- Updates after every new detection
- Up to 546 features per object



Alert spatial distribution (as of today) over Legacy Surveys DR10 image

ALeRCE public real-time products from stream data (as of today)

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Exploring the sky with ALeRCE: tips

- Start with small samples: 1 object, 10, 100, …
- For classifiers in general:
  - Use ALeRCE Explorer or jupyter notebook to select samples from the ALeRCE database
    - In case of query timeout problems, use cuts in first detection date
      - In Explorer: Discovery Date Filters -> Min MJD
      - In database: firstmjd in ‘object’ table)
      - If this still does not help, make your query smaller and divide in chunks
    - Use Explorer to inspect your sample
      - Supports both individual objects and lists up to 200 objects
        - Note that finding several rows for a given object mean several classifier versions
      - Check light curve from both alerts (ZTF, LSST) and a ZTF DR
      - Check ancillary data (‘other archives’ button)